VL53L1X Multi-Sensor Setup Guide

1. Hardware Connection Mapping

Each sensor requires power (3.3V), Ground, and a dedicated pair of SDA/SCL pins. For this 4-sensor configuration, the default I2C bus is used plus three additional software-defined buses.

Sensor	Component	Pi's Physical Pin	RPi Function
All Sensors	VDD (Pin 5)	Pin 1	3.3V Power
All Sensors	GND (Pin 6)	Any GND Pin	Ground
Sensor 1	SDA / SCL	Pin 3 / Pin 5	I2C1 (Standard)
Sensor 2	SDA / SCL	Pin 7 / Pin 29	I2C3 (Modified)
Sensor 3	SDA / SCL	Pin 32 / Pin 33	I2C5 (Modified)
Sensor 4	SDA / SCL	Pin 15 / Pin 16	I2C6 (Modified)

Note: On the sensor PCB, the square pin indicates PIN 1 in the datasheet.

2. Enabling Additional I2C Buses

To use four sensors, modify the system configuration to enable extra I2C interfaces:

- 1. Open terminal and run: sudo nano /boot/firmware/config.txt
- 2. Add these lines to the end of the file:

```
# Enable additional I2C buses (Avoiding SPI pins)
dtoverlay=i2c3,pins_4_5
dtoverlay=i2c5,pins_12_13
dtoverlay=i2c6,pins_22_23
```

- 3. Reboot the Raspberry Pi.

3. Verification

Check for address 0x29 on each bus using:

- sudo i2cdetect -y 1
- sudo i2cdetect -y 3
- sudo i2cdetect -y 5

- sudo i2cdetect -y 6

4. Software Execution

Clone the repository and run the multi-sensor test:

```
git clone https://github.com/slaghuis/VL53L1X-Raspberry-Pi.git
cd ~/your /directory /whatever
g++ test_multi.cpp vl53l1x_multi.cpp -o test_multi
./test_multi
```